

EXAM POV NOTES (Q&A FORMAT)

10 Marks Questions (11 points each)

Q1: Elaborate on the key foundational areas that underpin the development of Artificial Intelligence systems, detailing their primary contributions.

The development of intelligent systems in AI is built upon several key foundational areas, each contributing distinct insights and tools:

- **Philosophy:** This area addresses fundamental questions about the nature of knowledge, reasoning, and intelligence.
- **Ethical Considerations:** A crucial aspect originating from philosophy, guiding the responsible development and deployment of AI.
- **Mathematics:** Provides essential logical and quantitative frameworks, including logic, probability theory, optimisation, and linear algebra, which are vital for AI algorithms and models.
- **Computer Science:** Contributes the algorithmic and computational tools necessary for building AI systems, such as algorithms, data structures, complexity theory, and programming languages.
- **Cognitive Science:** Offers an understanding of human cognition, including models of perception, memory, and decision-making, which can inspire AI system design.
- **Linguistics:** Is crucial for natural language processing and achieving semantic understanding within AI systems.
- **Neuroscience:** Provides insights into brain structure and function, serving as an inspiration for neural network architectures.
- **Statistics:** Supports data analysis and interpretation, hypothesis testing, and model validation, which are critical for learning from data in AI.
- **Psychology:** Contributes theories of learning and development, as well as an understanding of human behaviour and decision-making, relevant to designing intelligent agents.
- **Control Theory:** Focuses on designing systems that can regulate their own behaviour, which is applicable to autonomous AI systems.
- **Information Theory:** Deals with quantifying information and its transmission, underpinning data compression and communication within AI systems.

2 Marks Questions (2 points each)

Q1.1: What is the primary goal of the "acting rationally" approach in AI?

- The primary goal of the "acting rationally" approach in AI is to **make decisions that maximise expected utility**.
- This perspective focuses on developing systems that take actions to achieve the best outcome based on available information and defined goals.

Q2: Discuss the current "State-of-the-Art" in Artificial Intelligence, highlighting various advanced technologies and emerging research areas.

The current "State-of-the-Art" in Artificial Intelligence is dynamic and encompasses a wide range of advanced technologies and emerging research fields:

- **Large Language Models (LLMs):** Models like **GPT-4** demonstrate impressive natural language understanding and generation capabilities, finding applications in text generation, translation, and question-answering.
- **Computer Vision:** Features advanced object detection, image segmentation, and generative models such as **DALL-E** and **Stable Diffusion**, which can create images from text descriptions.
- **Reinforcement Learning:** AI agents are mastering complex games (e.g., **AlphaGo**, **AlphaStar**) and find significant applications in robotics and autonomous systems.
- **Multimodal AI:** Involves systems that can process and generate multiple types of data, including text, images, and audio, with examples like GPT-4 demonstrating image understanding capabilities.
- **AI for Scientific Discovery:** Includes breakthroughs such as **AlphaFold's** protein structure prediction, along with AI-assisted drug discovery and materials science advancements.
- **Autonomous Vehicles:** Continues to see advancements in self-driving technology and the integration of AI into various transportation systems.
- **AI in Healthcare:** Utilises diagnostic tools based on machine learning and offers personalized treatment recommendations.
- **Explainable AI (XAI):** Focuses on developing methods to make AI decision-making processes more transparent and interpretable for users.
- **AI Ethics and Fairness:** Represents an ongoing research area dedicated to reducing bias and ensuring the ethical deployment of AI systems.
- **Quantum AI:** Explores the intersection of quantum computing and AI algorithms, with the potential to achieve new computational capabilities.
- **Neuromorphic Computing:** Aims to develop hardware that mimics brain structure for more efficient AI processing.

5 Marks Questions (6 points each)

Q3: Define Artificial Intelligence, explain its core aim, and describe its main categories with examples.

- **Definition of AI:** AI refers to the development of computer systems that can perform tasks that typically require human intelligence.
- **Core Aim of AI:** The field of AI aims to create machines that can mimic or even surpass human cognitive abilities in specific domains.
- **Examples of AI Tasks:** AI systems are capable of tasks such as learning and problem-solving, understanding natural language, recognising speech and visual patterns, and making decisions and predictions.
- **Category 1: Narrow AI:** Also known as Weak AI, these systems are designed for specific tasks.
- **Example of Narrow AI:** Virtual assistants (like Siri or Alexa) and image recognition systems are common examples of Narrow AI.
- **Category 2: General AI:** This is a hypothetical concept, referring to systems with human-like intelligence across various domains, theoretically capable of performing any intellectual task that a human can.

Q4: Explain the difference between "thinking humanly" and "acting rationally" approaches in AI, providing key aspects for each.

The "thinking humanly" and "acting rationally" approaches represent two fundamental philosophies in Artificial Intelligence:

- **Thinking Humanly Approach:** This approach refers to designing AI systems that mimic or emulate human cognitive processes.
- **Key Aspects of Thinking Humanly:** It focuses on aspects such as cognitive modelling, psychological basis, human-like reasoning, emotional intelligence, learning and adaptation, perception, and natural language processing.
- **Example for Thinking Humanly:** An AI system attempting to emulate human thinking might involve cognitive architectures that simulate how humans solve problems, or systems focused on understanding emotional cues.
- **Acting Rationally Approach:** This approach refers to developing systems that take actions to achieve the best outcome or maximise their expected utility.
- **Key Aspects of Acting Rationally:** This perspective focuses on goal-oriented behaviour, rational decision-making, optimisation, adaptability, resource management, handling uncertainty, and long-term planning.
- **Example for Acting Rationally:** An AI system like a chess-playing computer that calculates the optimal moves to win, or a navigation system that finds the fastest route, exemplifies acting rationally.

Q5: Outline six significant milestones or periods in the "History of AI," including the concept of "AI Winters."

The history of AI is marked by periods of great advancement and periods of reduced interest, often referred to as "AI Winters":

- **Early Foundations (1940s-1950s):** Key events include **Warren McCulloch and Walter Pitts** proposing the first artificial neuron in 1943, and **Alan Turing** introducing the Turing Test in 1950.
- **The Golden Years (1950s-1960s):** This period saw the **Dartmouth Conference** in 1956 coin the term "Artificial Intelligence," alongside the development of early AI programs like Logic Theorist and the LISP programming language.
- **The First AI Winter (1970s):** This was a period characterised by the limitations of existing AI approaches becoming apparent, leading to a significant decline in funding and general interest in the field.
- **Expert Systems and Revival (1980s):** This decade saw the rise of expert systems in commercial applications and a renewed interest sparked by **Japan's Fifth Generation Computer Project**.
- **Second AI Winter (Late 1980s-Early 1990s):** Following the expert systems boom, another period of reduced funding and interest occurred as expert systems proved limited and expensive.
- **Emergence of Machine Learning (1990s-2000s):** This era saw a shift in focus to data-driven approaches, the development of probabilistic methods and neural networks, and notable AI successes in games, such as **Deep Blue defeating the world chess champion in 1997**.